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Data Science

AgriAid+: AI-Powered Smart Farming & Climate Resilience Platform

Group No. 1

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Abstract

Agriculture in India faces challenges like unpredictable climate, limited scientific guidance, language barriers, and market inefficiencies that impact farmer profitability. To address these, **AgriAid+** introduces an AI-powered smart farming and climate resilience platform with four modules: a Crop & Fertilizer Recommendation Engine using soil, weather, and regional data; a Multilingual Farmer Advisory Chatbot for local-language support; Weather & Market Trend Analysis for better planning; and a Farmer-Centric Marketplace for direct trading. By combining AI, NLP, and real-time analytics, AgriAid+ empowers farmers with insights, improves productivity, ensures fair pricing, and promotes sustainable, climate-resilient agriculture.

Sustainable Development Goals(SDG) mapped

SDG no. : 13

SDG description: Climate Action

How it is mapped:
Weather forecasting and climate-aware crop planning help farmers adapt to changing climatic conditions, reducing crop loss and supporting climate resilience in agriculture.

Introduction

Agriculture is the backbone of India's economy, yet farmers face challenges like unpredictable weather, lack of data-driven crop and fertilizer guidance, market inefficiencies, and language barriers. Existing tools such as weather apps and government portals are often fragmented, impersonal, and not available in regional languages. This highlights the need for a unified AI-powered platform that delivers real-time, localized, and practical solutions to help farmers make better decisions, improve productivity, and secure fair incomes.

Objectives

•To develop a personalized crop and fertilizer recommendation system using soil data, leveraging advanced machine learning algorithms such as XGBoost and LightGBM.

•To design a multilingual farmer advisory chatbot that enables seamless communication and support in local languages.

•To integrate weather forecasting and market price analytics for better crop planning .

•To build a farmer-centric digital marketplace for direct trading of produce and agricultural equipment.

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Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	Advancing crop recommendation system with supervised machine learning and explainable artificial intelligence	Sourabh Shastri, Sachin Kumar, Vibhakar Mansotra & Rohit Salgotra	2025	Supervised machine learning (Gradient Boosting, KNN, DT, RF, NB, SVM, LR, NN, LDA, QDA) and Explainable AI (LIME)	The model's excellent performance on the limited, single Kaggle dataset may not generalize well to varied real-world agricultural conditions.
2	Intent Aware Context Retrieval for Multi-Turn Agricultural Question Answering	Abhay Vijayvargia, Ajay Nagpal, Kundeshwar Pundalik, Smita Gautam, Atharva Savarkar, Viraj Thakur, Pankaj Singh, Rohit Saluja, Ganesh Ramakrishnan	2025	IFT model refined with fine-tuning on Indian agricultural knowledge, multi-turn conversation flow, Retrieval-Augmented Generation (RAG)	The model is currently specialized in only two crops: onions and grapes. Expanding support to a wider variety of crops requires significant dataset expansion and re-engineering of the intent-slot mapping and RAG module.
3	Implementation, Impact, and Challenges of eNAM in Agricultural Markets – A Systematic Literature Review (SLR)	Sushil Chauhan & Shivali Rathore	2025	Systematic Literature Review (SLR)	The review's scope is limited to English, peer-reviewed journal articles from 2016-2024, potentially omitting other relevant research.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
4	Machine Learning in Sustainable Agriculture: Systematic Review and Research Perspectives	Juan Botero-Valencia, Vanessa García-Pineda, Alejandro Valencia-Arias, Jackeline Valencia, Erick Reyes-Vera, Mateo Mejia-Herrera, Ruber Hernández-García	2025	PRISMA-based Systematic Review; Bibliometric Analysis of ML in agriculture (2007–2025, 124 studies)	Reliance on selected databases and limited bibliometric tools may introduce biases and restrict comprehensiveness.
5	Digital Agro-Advisory Tools	Kingsley Ofosu-Ampong, Wuletawu Abera, Tewodros Mesfin	2025	Review of 19 platforms; adoption & behavioural analysis.	Relies on online use; low adoption in rural areas.
6	Hybrid SV-PSO-BiLSTM	Satish Kumar, Amit Verma, Pankaj Kumar, Rakesh Majhi	2025	Hybrid model for agri-price prediction with PSO & BiLSTM.	High complexity & time; limited to prices.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
7	Agro-Digital Tool Adoption	Kingsley Ofosu-Ampong, Tsion Abate, Wuletawu Abera	2025	Categorised tools by adoption (low/med/high).	Limited data; findings not generalisable.
8	ML for Climate-Smart Agriculture	R. Kumar, S. Singh, P. Sharma	2024	Conceptual ML approach for CSA adoption & productivity.	Lacks long-term local climate guidance.
9	Data-driven irrigation optimization	Imade	2024	RF with soil & weather data cut water use by ~25%	Costly sensors, low generalizability.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
10	ML for yield prediction in smart farming	Abourabia Imade, Ounacer Soumaya, Elghoumari Mohammed yassine, Azzouazi Mohamed	2024	Compared regressors; AdaBoost lowest error, RF R^2≈0.89; DL image models ≈99%.	Region specific , data quality issues
11	Smart agriculture: ML & DL for drought stress	TariqAli1, Saif Ur Rehman, Shamshair Ali, Khalid Mahmood, Silvia Aparicio Obregon, Rubén Calderón Iglesias, Tahir Khurshaid & Imran Ashraf	2024	Used SVM, GB, RNN, LSTM on plant protein features; accuracy up to 97%.	Limited field validation, high computation.
12	Predicting climate smart agriculture (CSA) practices using machine learning: A prime exploratory survey	Freddy Noma , Suresh Babu	2024	Optimized Gradient Boosting ML algorithm trained and tested using household-level data from Rakai district in Uganda, framed within climate adaptation analytical frameworks.	The model achieved only up to 60% accuracy, indicating a significant portion (40%) of misclassifications, which the authors suggest could be improved with deep-learning algorithms in future research.

Research Gap(Limitations of existing systems)

- Most chatbots speak in only one or two languages, leaving out millions of farmers who rely on local dialects to access knowledge.
- Platforms like eNAM still struggle with trust — without ratings or feedback, farmers hesitate to depend on them for critical market decisions.
- Small and noisy datasets weaken the reliability of predictions, often making recommendations less useful in real-world fields.
- Current systems often ignore weather and climate factors, even though these are the biggest forces shaping agriculture today.
- Advanced ML models exist, but they demand heavy computation, keeping powerful solutions out of reach for everyday users.
- Long-term climate data is missing, leaving systems unprepared to guide farmers in adapting to changing weather patterns.

Problem Definition

- **Challenges Faced by Farmers** – Indian farmers struggle with lack of data-driven crop/fertilizer decisions, unpredictable climate risks, limited access to real-time weather/market data, and language barriers in understanding advisories and schemes.

- **Limitations of Current Solutions** – Existing platforms and government initiatives are fragmented, offer minimal personalization, and lack integration of essential agricultural services.

- **Proposed System: AgriAid+** – A unified AI-powered platform integrating crop and fertilizer recommendations, multilingual advisory chatbot, weather forecasting, market price analytics, and a farmer-centric marketplace.

Scope

- To develop a crop and fertilizer recommendation system that uses soil data, weather data, and agricultural datasets to provide personalized guidance.

- To build a multilingual farmer advisory chatbot that enables farmers to interact in local languages and access information on crop care, government schemes, and best practices.

- To integrate weather forecasting and market price trend analysis for supporting better crop planning, harvesting, and selling strategies.

- To design a farmer-centric marketplace module that allows direct trading of produce and agricultural equipment, ensuring transparency and fair pricing.

- To provide a unified AI-powered platform that combines decision support, real-time insights, and marketplace services to empower farmers and promote sustainable agriculture.

Technological Stack

Crop Health Prediction (XGBoost)

- **Stack:** Python, XGBoost, Pandas, Scikit-learn
- **Method:** Trained XGBoost on soil, weather, and sensor data to classify crop health.
- **Result:** Accuracy ~93%; detects early signs of nutrient deficiency and stress.

Fertilizer Recommendation (XGBoost)

- **Stack:** Python, XGBoost, NumPy, Scikit-learn
- **Method:** Predicts optimal fertilizer type & quantity per crop based on soil & yield data.
- **Result:** Accuracy ~96%; improves input efficiency and yield.

Nakshatra-Based Crop Prediction

- **Stack:** Python, PyEphem, Pandas, NumPy
- **Method:** Calculates Moon longitude using PyEphem to determine Nakshatra for a given date. Maps Nakshatra to fertility rating and recommends suitable crops based on Vedic principles.
- **Result:** Provides crop recommendations with alignment to Moon phases; helps optimize sowing and planting schedules.

Proposed system architecture/Working/Module Diagram

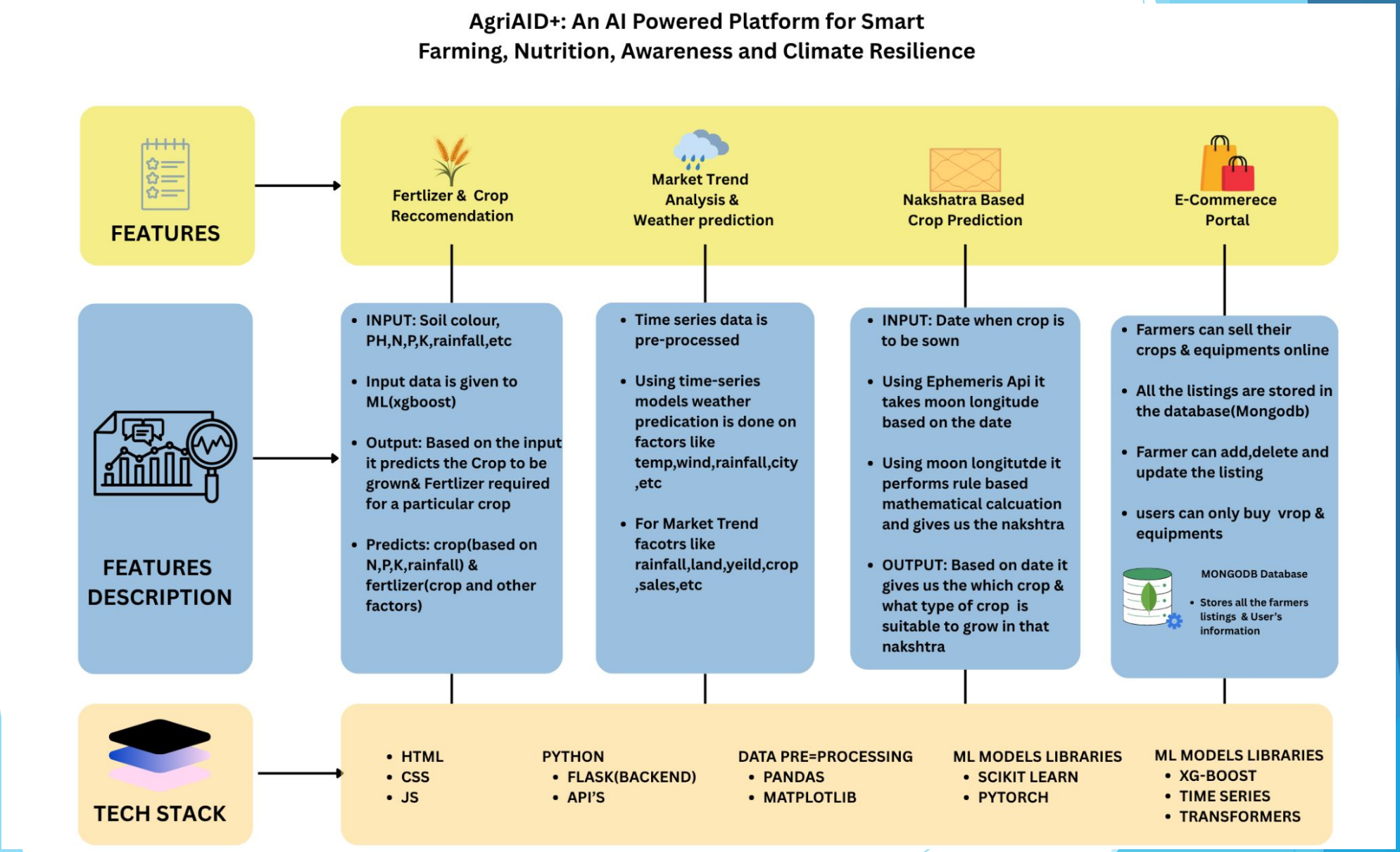

Fig 1: Architecture design for Agridaid+

Prototype Design Demonstration

Fig 1 :Home Page of AgriGrid

Prototype Design Demonstration

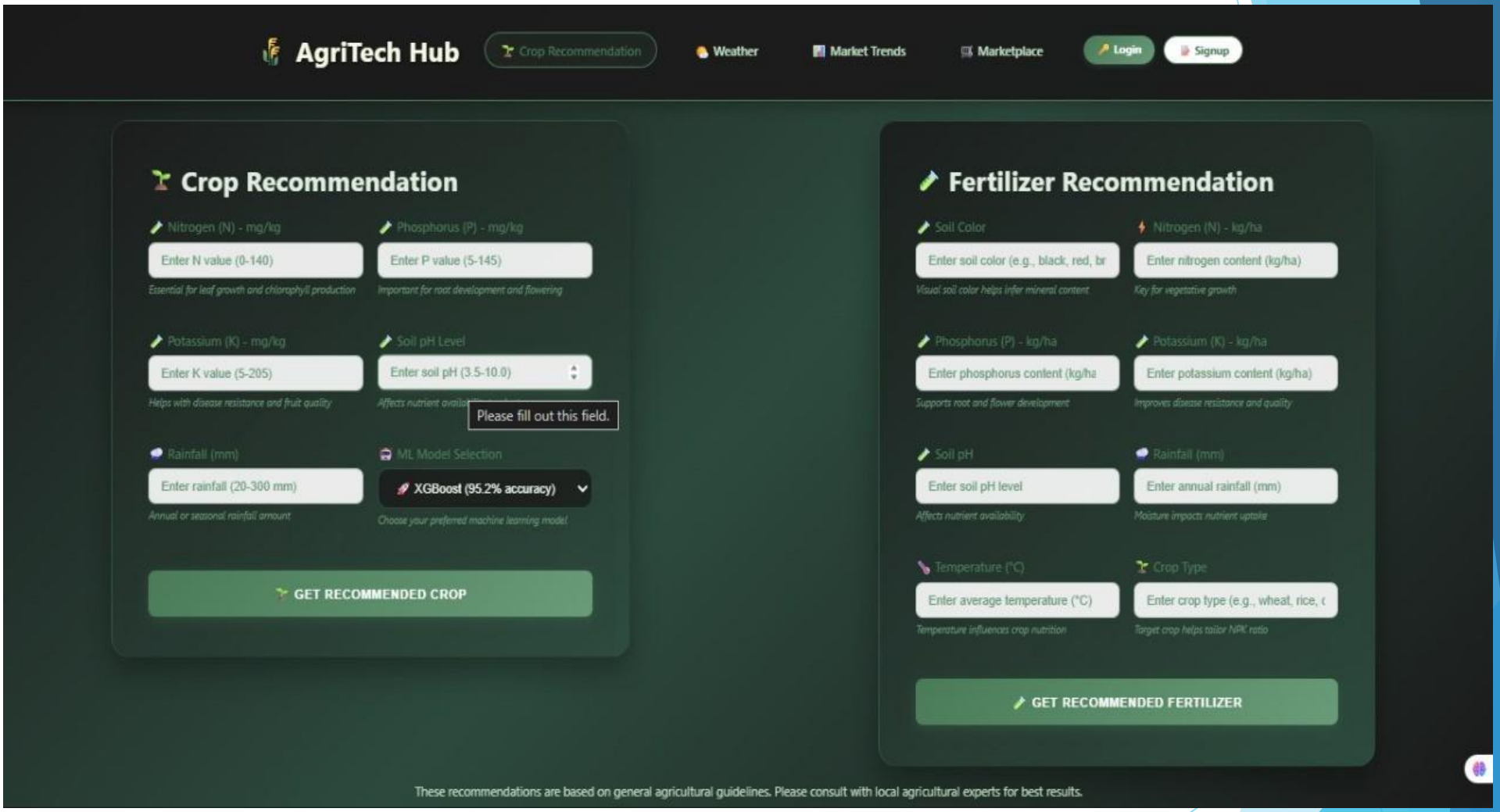

Fig 2 :Crop Recommendation & Fertilizer Recommendation

Prototype Design Demonstration

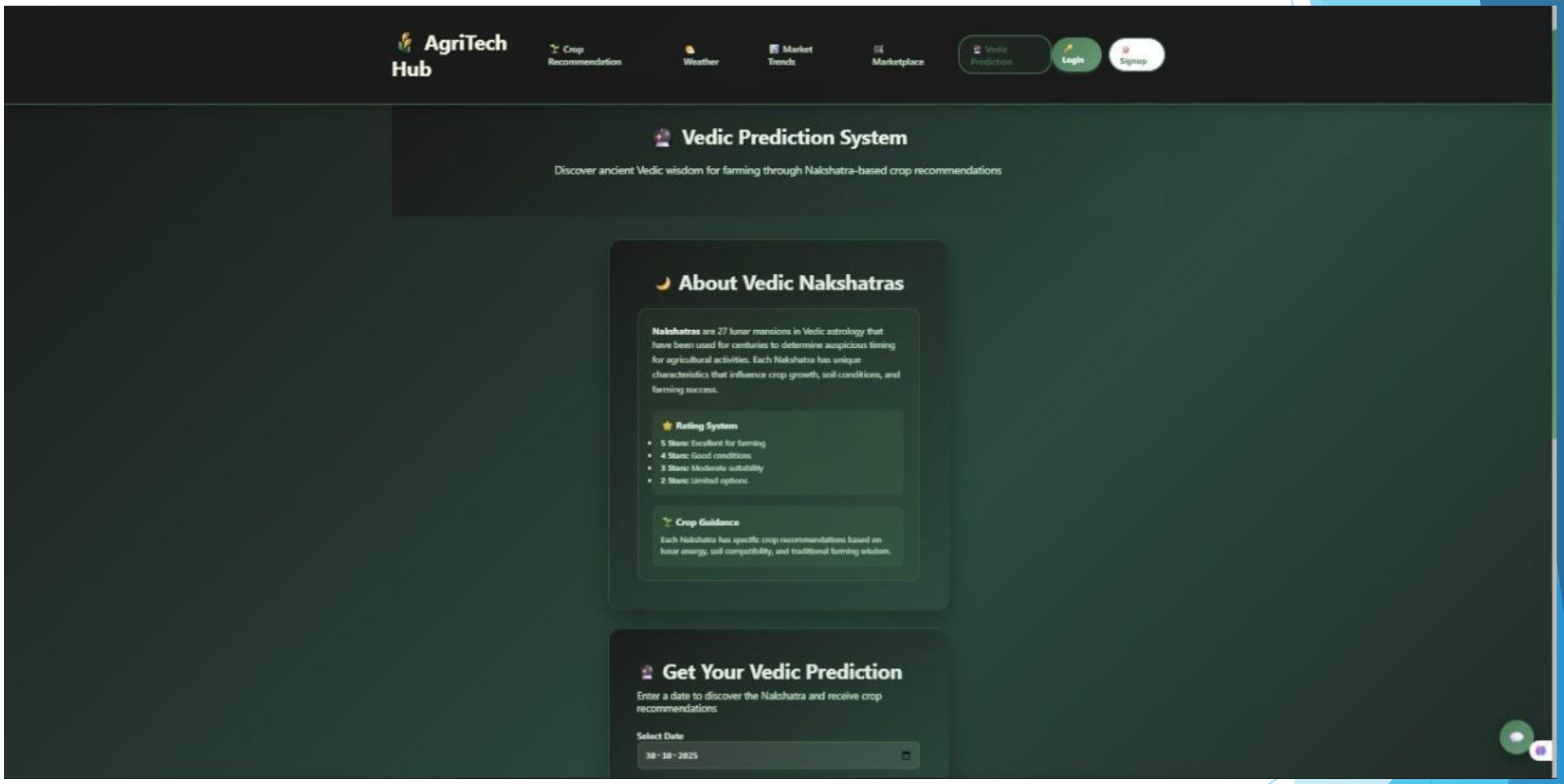

Fig 3 :Vedic Nakshatra Prediction

Implementation Status

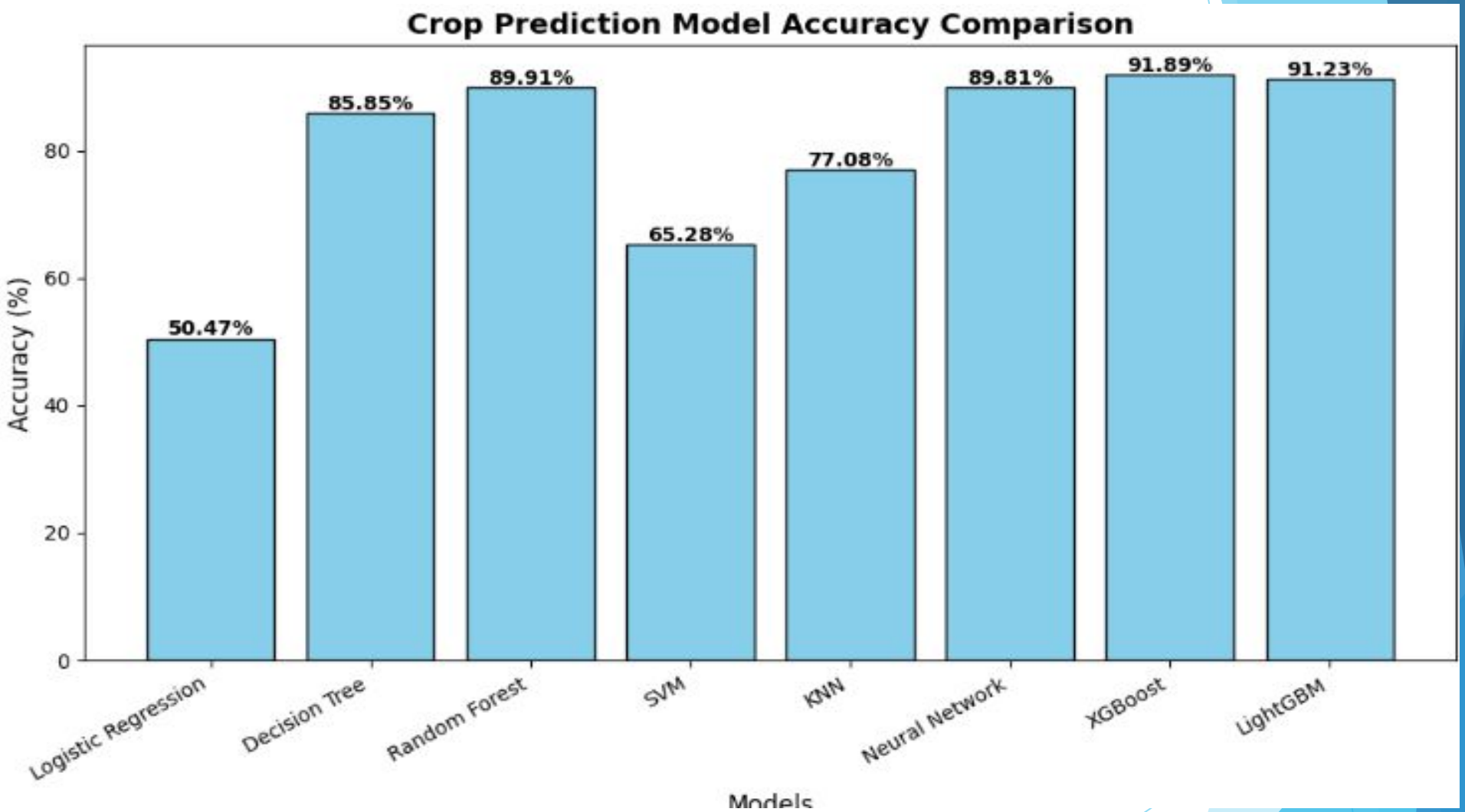

Fig 2 :Comparative Study of ML Algorithms for Crop Classification

Implementation Status

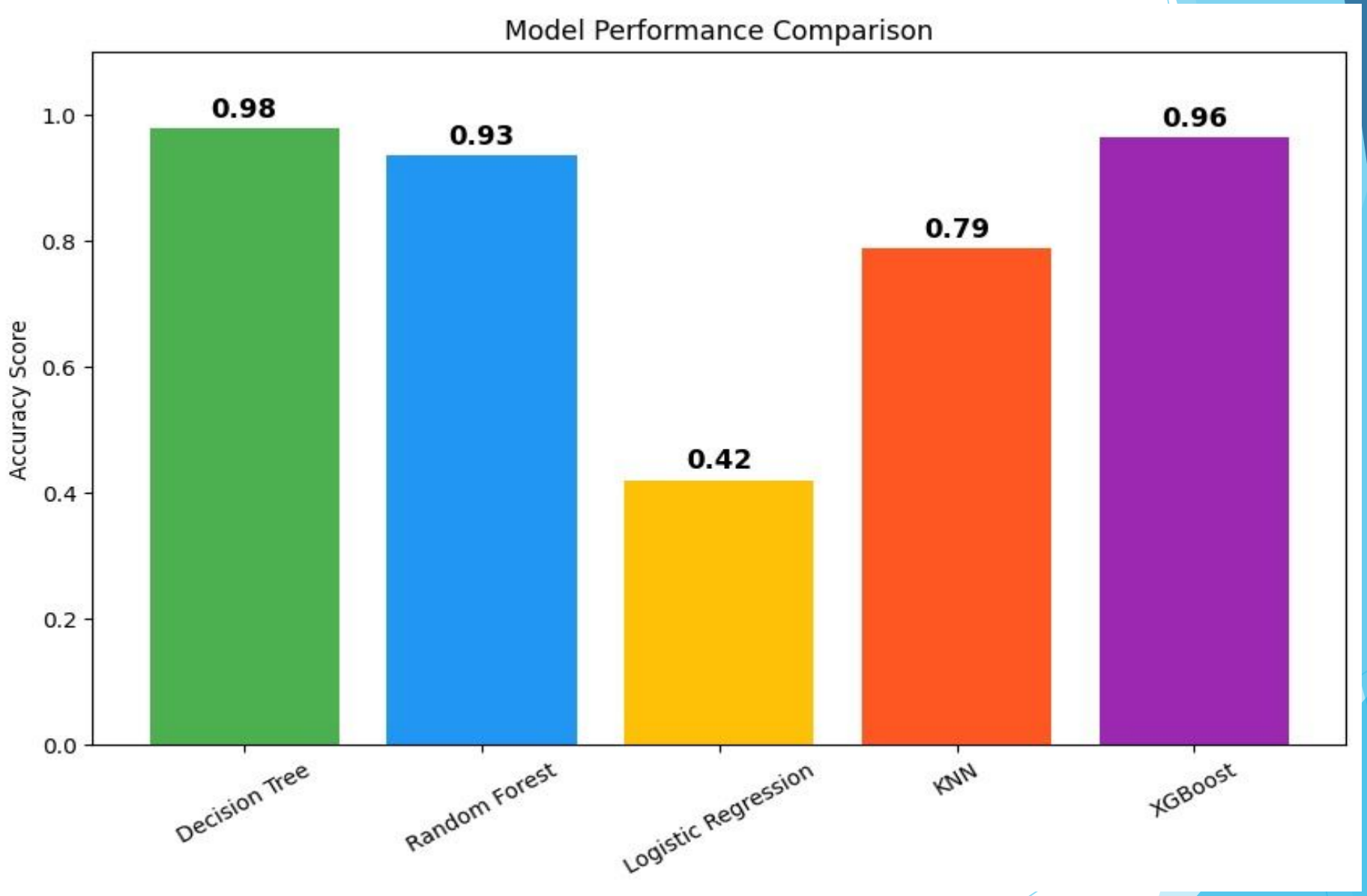

Fig 3 :Comparative Study of ML Algorithms for Fertilizer Recommendation

Implementation Status

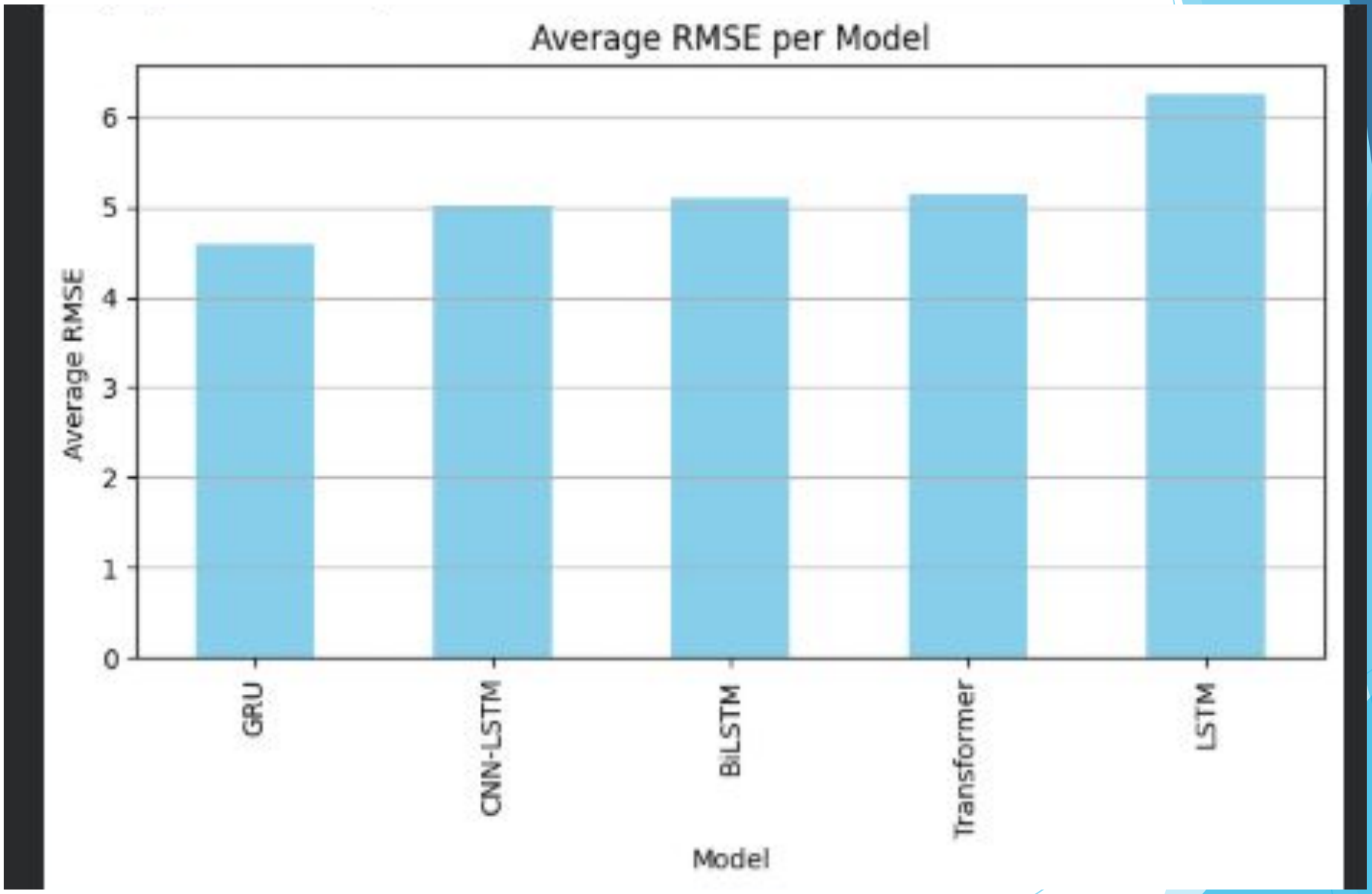

Fig 3 :Comparative Study of ML Algorithms for Weather Prediction

Review Suggestions (Given in Last meeting)

- **Suggestion:** Clarify how the user will provide soil input (pH level, phosphorus, nitrogen, etc.) for crop recommendation.

- **Action Taken:** Integrated an intelligent agent to handle soil data input and guide users in entering required parameters for accurate crop recommendations.

- **Suggestion:** Limit the agricultural area to Maharashtra instead of covering the entire country, and focus on predicting only 2–3 crops rather than multiple crops across different regions.

- **Action Taken:** The scope of the project was restricted to Maharashtra to improve regional relevance. For market trend analysis, only two crops potato and onion were selected, and data was collected from five major cities in Maharashtra, enabling more focused and reliable predictions.

Research paper Status

- The research paper is around 90% complete. Similarity has been checked and is within acceptable limits. The suggested changes are being incorporated.

Patent/Copyright filing Status

- The copyright process has not been initiated yet. We will initiate the process shortly.

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[14] Devendra Dahiphale, Pratik Shinde, Koninika Patil and Vijay Dahiphale, **Smart Farming: Crop Recommendation using Machine Learning with Challenges and Future Ideas**, 2023.

[15] Hitiksha Karre, C. Prakash and P. Balamurugan, **Soil Classification & Crop Recommendation**, 2020.

Thank You...!!